

Code:-

```
1 #include <stdio.h>
2 #include <stdlib.h>
3 #include <unistd.h>
4 #include <sys/types.h>
5 #include <sys/wait.h>
6 #include <string.h>
7
8 int main()
9 {
10     char command[100];
11     char *args[20];
12     char *token;
13     int i = 0;
14
15     printf("Enter a Linux command: ");
16     fgets(command, sizeof(command), stdin);
17
18     // Remove newline
19     command[strcspn(command, "\n")] = '\0';
20
21     // Split command into arguments
22     token = strtok(command, " ");
23
24     while (token != NULL && i < 19)
25     {
26         args[i++] = token;
27         token = strtok(NULL, " ");
28     }
29
30     args[i] = NULL;
31
32     pid_t pid = fork();
33
34     if (pid < 0)
35     {
36         perror("fork failed");
37         return 1;
38     }
39
40     if (pid == 0)
41     {
42         // Child process
43         printf("\nChild Process\n");
44         printf("Child PID : %d\n", getpid());
45         printf("Parent PID : %d\n", getppid());
46
47         execvp(args[0], args);
48
49         // Executes only if execvp fails
50         perror("execvp failed");
51         exit(1);
52     }
53     else
54     {
55         // Parent process
56         printf("\nParent Process\n");
57         printf("Parent PID : %d\n", getpid());
58         printf("Child PID : %d\n", pid);
59
60         wait(NULL);
61
62         printf("\nChild process completed.\n");
63     }
64
65     return 0;
66 }
```

Output:

```
saishurendra@saishurendra:~$ nano Experiment1.c
saishurendra@saishurendra:~$ gcc Experiment1.c -o Experiment1
saishurendra@saishurendra:~$ ./Experiment1
Enter a Linux command: ls

Parent Process
Parent PID : 634
Child Process
Child PID : 635
Child PID : 635
Parent PID : 634
Client      Experiment3.c  Operating   Practical11.c Practical15.c Practical18.c Practical4.c Practical7.c Program1.c Program5.c Serve Skill12.c and snap
Client.c      Experiment4.c  Practical15.c Practical12.c Practical15.c Practical19.c Practical4.c Practical8.c Program1.c Program5.c Serve.c Skill13.c expl
Experiment1 Experiment4.c  Practical11.c Practical12.c Practical16.c Practical19.c Practical5.c Practical8.c Program1.c Program5.c ProgrammingPractical Signal Skill13.c expl.c
Experiment1.c FCFS Practical11.c Practical13.c Practical16.c Practical2.c Practical5.c Practical9.c Program1.c Program5.c RoundRobin.c Signal.c Skill14.c hello
Experiment2.c FCFS Practical10.c Practical13.c Practical17.c Practical2.c Practical6.c Practical9.c Program2.c Program2.c RoundRobin.c Skill1 Skill14.c hello.c
Experiment2.c OS Practical10.c Practical14.c Practical17.c Practical3.c Practical6.c Priority.c Program3.c SJF Skill11.c SurendraDesktopOperating info.txt
Experiment3.c OSCP Practical11.c Practical14.c Practical18.c Practical3.c Practical7.c Priority.c Program4.c SJF.c Skill12 Systems hello.txt
Child process completed.
```